

Mark Opfell

Skills and Exposure

Standards	IRIG, CCSDS, DVB-S2, ITU, FCC
RF Tools	SDR, VNA, GNU Radio, VSA
General Software Tools	Python, Git*, Linux, Bash, AWS EC2
Python Libraries	NumPy, Matplotlib, Scapy
Networking	Ethernet, UDP/IP, Wireshark, iPerf
FPGA	Xilinx, PYNQ, Vivado
Significant Volcano Ascents	Rainier, Baker, Adams

Experience

Job Title	Senior RF and Telemetry Engineer	
Employer	Relativity	Long Beach, CA
Period	August 2025 – Present	
Networked, programmed, built, and tested RF HITL rack routing S-band telemetry-over-IP from space rocket UDP/IP multicast network traffic to radiated PSK waveform received by the ground receiver. Wrote, and tested high-level code generating HDL for a pseudo-randomization algorithm proving basic spectral compliance. Hands-on debugged a Xilinx FPGA software defined transmitter’s digital frame synchronizer fixing major errors in high level HDL.		
Job Title	RF Communications System Engineer	
Employer	Amazon: Kuiper	Redmond, WA
Period	July 2024 – November 2024	
Developed and ran, over the air Ka-band MIMO phased array system test UDP/IP throughput tests on Xilinx Versal FPGAs for a LEO constellation.		
Job Title	Lead Communication Systems Engineer	
Employer	Albedo	Remote & Some Travel
Period	October 2021 – March 2024	
Built a large scale year-in-the-life Python link budget modeling satellite constellation average payload data throughput. Created, evaluated, and built space-to-ground digital communications payload (gigabit class) for high data rate visible and thermal sensors for VLEO satellites. Developed the telemetry and command data chain from modulated waveform to frames, packets, and connections. Analyzed and tested with: software defined AD9361-based packet radios, GNU radio, technical deep dives into CCSDS standards.		

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Job Title	Senior RF Systems Engineer	
Employer	BlackSky	Tukwillla, WA & Remote
Period	April 2019 – October 2021	

Created RF architecture diagrams, link budgets, test plans, and ran hands-on troubleshooting. Collaborated with customers and suppliers to design, manufacture, test, launch, and operate X (payload), S (TT&C), GPS, and UHF-band space-based software defined radios linked to ground stations enabled by AWS and the KSAT Lite ground station network.

Collaboratively designed, simulated, sourced, advised layout, and validated: parts, mixed signal PCB, connectors, cabling, and enclosure for a GPS RF system self-compatibility filter. Multiple spacecraft successful in-orbit operation.

Job Title	RF Systems Engineer	
Employer	Kymeta	Redmond, WA
Period	February 2018 – March 2019	

Developed and executed over-the-air combined OSI application, transport, network, and physical layer level test cases for a mobile Azure cloud connected MIMO Ku-band terminal with software defined phased array flat panel antennas and a DVB-S2 satellite modem

Job Title	Senior RF Systems Engineer	
Employer	Maxar	Mountain View, CA
Period	September 2013 – January 2018	

Wrote specifications, triaged vendors, reviewed test data collateral, and directed the installation, unit level and system level tests of the following passive and active RF units: diplexer, waveguide, directional coupler, band pass filter, low noise amplifier, downconverter, high power load, circulator, coaxial cable, master reference oscillator, and synthesizer.

Developed Python analysis tool from scratch to model complex amplitude and time delay of 10,000+ RF units for ground-based beam-forming.

Education & Certifications

Degree	Bachelor of Science in Electrical Engineering
University	University of California, Davis
Period	2009 – 2012

Certification	Network Technician
Organization	Cisco
Period	2024

Certification	Apprentice Alpine Mountain Guide
Organization	American Mountain Guide Association
Period	2023

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